

Numerical Computing Assignment 2

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Subject: Numerical Computing Assignment 2

Algorithm

ForwardDifference(f, x, h)

```
{  
    f_x = f(x)  
    f_x_plus = f(x + h)  
  
    derivative = (f_x_plus - f_x) / h  
  
    RETURN derivative  
}
```

BackwardDifference(f, x, h)

```
{  
    f_x = f(x)  
    f_x_minus = f(x - h)  
  
    derivative = (f_x - f_x_minus) / h  
  
    RETURN derivative  
}
```

CentralDifference(f, x, h)

```
{  
    f_x_plus = f(x + h)  
    f_x_minus = f(x - h)  
  
    derivative = (f_x_plus - f_x_minus) / (2 * h)  
  
    RETURN derivative  
}
```

CompareDifferentiationMethods()

```
{  
    FOR each function f IN F DO  
    {  
        exact = AnalyticalDerivative(f, x0)  
  
        FOR each step size h IN H DO  
        {  
            forward = ForwardDifference(f, x0, h)  
            backward = BackwardDifference(f, x0, h)  
            central = CentralDifference(f, x0, h)
```

```

        error_forward = |exact - forward|
        error_backward = |exact - backward|
        error_central = |exact - central|

        Store (h, exact, forward, error_forward,
               backward, error_backward,
               central, error_central)
    }
}

Display results table

Plot log10(h) versus log10(error)

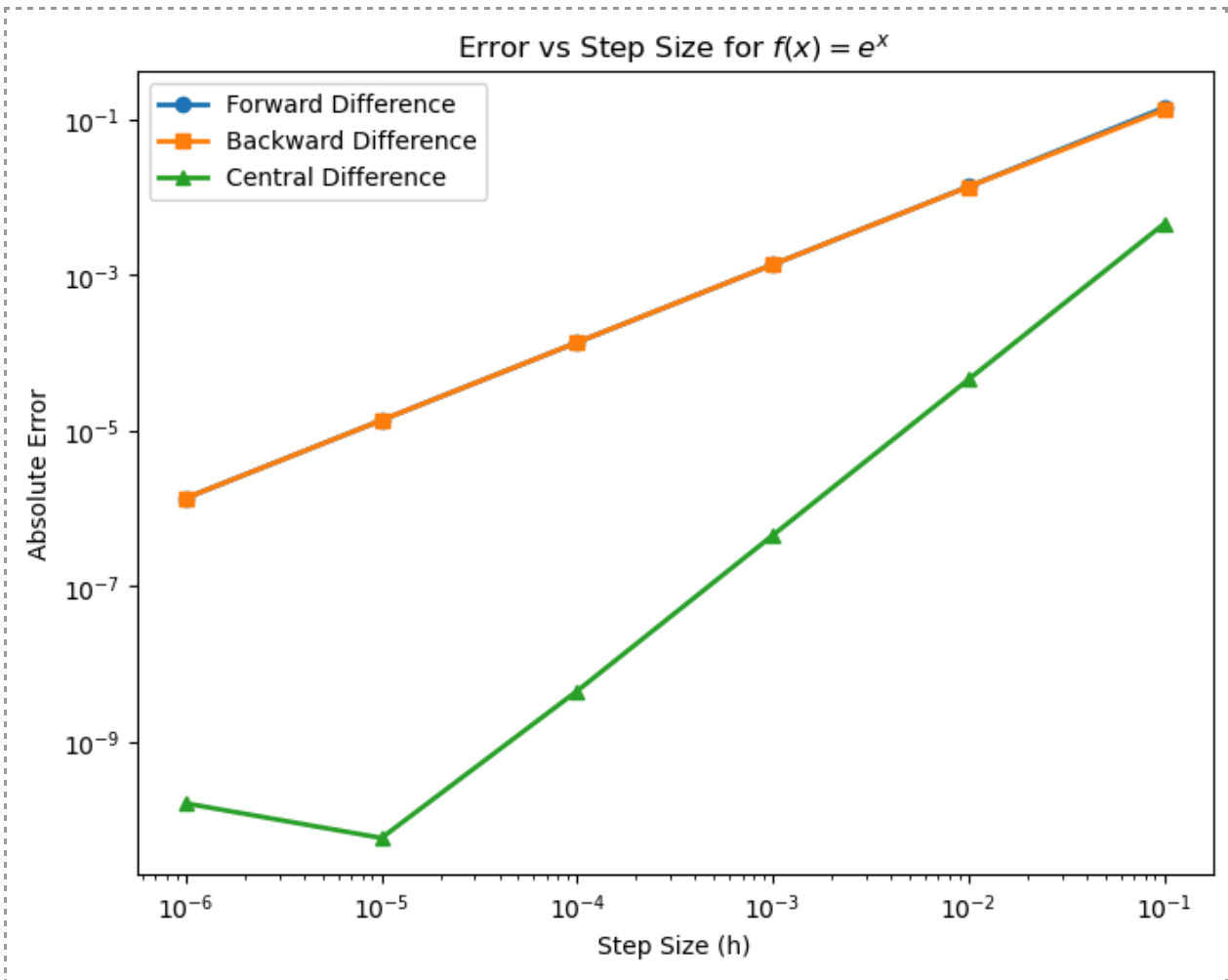
Compare convergence rates
Identify round-off error region
}

```

1st Equation: $f(x) = e^x$

$f'(1) \text{ exact} = e \approx 2.71828$

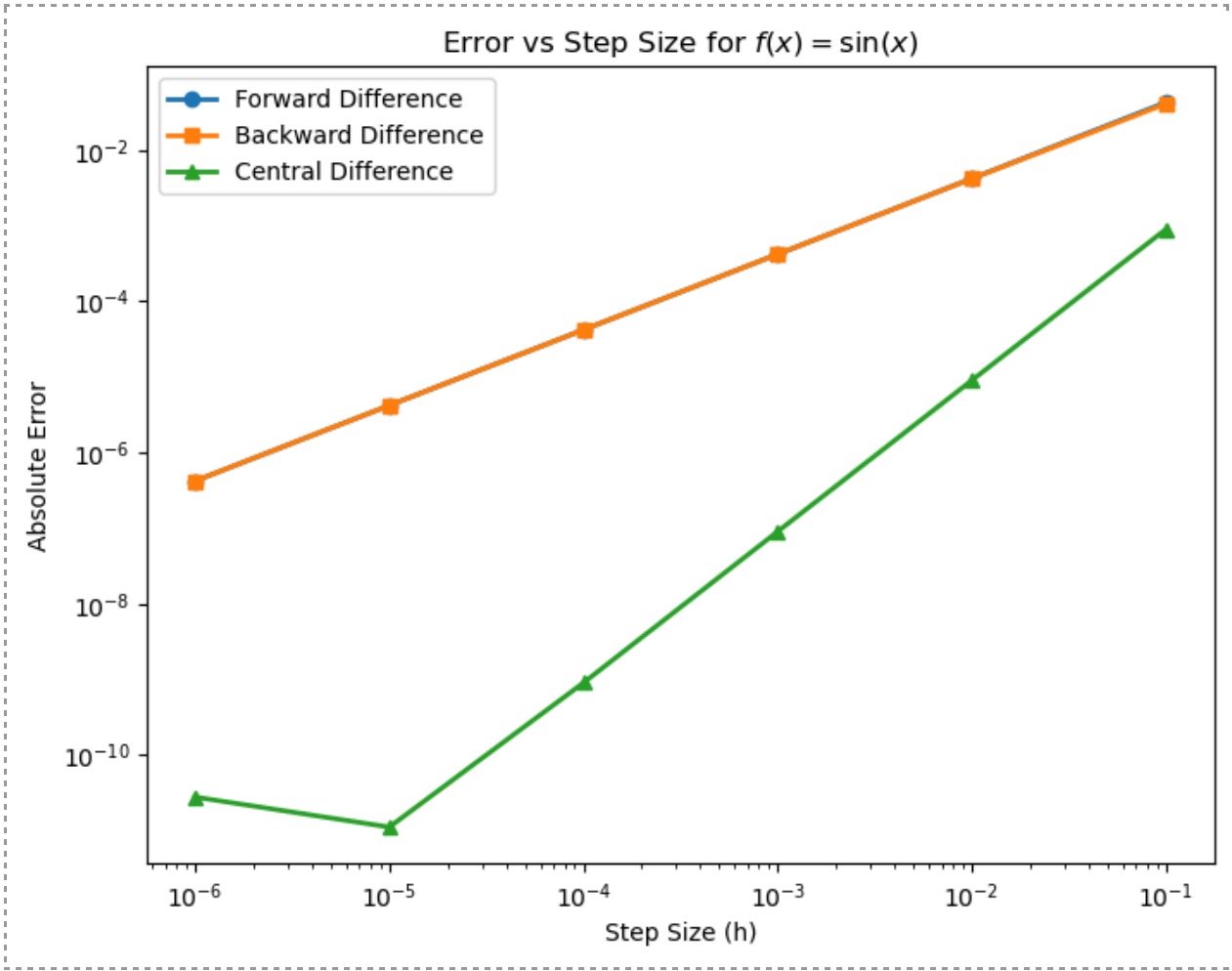
h	Exact f'(1)	Forward	Fwd Error	Backward	Bwd Error	Central	Central Error
1e-1	2.71828	2.85884	1.406e-1	2.58679	1.315e-1	2.72281	4.533e-3
1e-2	2.71828	2.73192	1.364e-2	2.70474	1.355e-2	2.71833	4.530e-5
1e-3	2.71828	2.71964	1.360e-3	2.71692	1.359e-3	2.71828	4.530e-7
1e-4	2.71828	2.71842	1.359e-4	2.71815	1.359e-4	2.71828	4.531e-9
1e-5	2.71828	2.71830	1.359e-5	2.71827	1.359e-5	2.71828	5.859e-11
1e-6	2.71828	2.71828	1.359e-6	2.71828	1.359e-6	2.71828	1.635e-10



2nd Equation: $f(x) = \sin(x)$

$f'(1) \text{ exact} = \cos(1) \approx 0.540302$

h	Exact f'(1)	Forward	Fwd Error	Backward	Bwd Error	Central	Central Error
1e-1	0.540302	0.497364	4.294e-2	0.581441	4.114e-2	0.539402	9.001e-4
1e-2	0.540302	0.536086	4.216e-3	0.544501	4.198e-3	0.540293	9.005e-6
1e-3	0.540302	0.539881	4.208e-4	0.540723	4.206e-4	0.540302	9.005e-8
1e-4	0.540302	0.540260	4.207e-5	0.540344	4.207e-5	0.540302	9.004e-10
1e-5	0.540302	0.540298	4.207e-6	0.540307	4.207e-6	0.540302	1.114e-11
1e-6	0.540302	0.540302	4.207e-7	0.540303	4.208e-7	0.540302	2.772e-11



3rd Equation: $f(x) = x^3 - 2x + 1$

$f'(1) \text{ exact} = 3(1)^2 - 2 = 1$

h	Exact f'(1)	Forward	Fwd Error	Backward	Bwd Error	Central	Central Error
1e-1	1.00000	1.31000	3.100e-1	0.710000	2.900e-1	1.01000	1.000e-2
1e-2	1.00000	1.03010	3.010e-2	0.970100	2.990e-2	1.00010	1.000e-4
1e-3	1.00000	1.00300	3.001e-3	0.997001	2.999e-3	1.00000	1.000e-6
1e-4	1.00000	1.00030	3.000e-4	0.999700	3.000e-4	1.00000	1.000e-8
1e-5	1.00000	1.00003	3.000e-5	0.999970	3.000e-5	1.00000	9.537e-11
1e-6	1.00000	1.00000	3.000e-6	0.999997	2.999e-6	1.00000	2.876e-11

